Table 1. Quality and cost under matched inference budgets. Success rate on four task suites and inference cost, compared within each token budget.

Method	Task success (%) ↑					Inference cost	
	Math n=250	**Code** n=200	**Docs** n=150	**Tools** n=100	**Avg.** macro	**Tokens** mean ↓	**Latency** ms ↓
16k-token budget							
Direct	65.6	61.5	73.3	54.0	63.6	**4,800**	**410**
Chain of thought	72.0	68.0	77.3	61.0	69.6	6,200	521
Search	78.8	74.0	82.0	65.0	75.0	10,150	811
Verifier	78.0	75.5	83.3	64.0	75.2	11,280	953
Adaptive routing	**80.4**	**77.5**	**85.3**	**69.0**	**78.1**	9,800	754
32k-token budget							
Chain of thought	73.6	70.5	80.7	64.0	72.2	**9,700**	**743**
Search	82.8	77.5	87.3	71.0	79.7	17,200	1,236
Verifier	83.2	79.0	88.7	72.0	80.7	18,400	1,403
Adaptive routing	**84.8**	**81.5**	**90.0**	**76.0**	**83.1**	15,100	1,128

Synthetic data. Rate = successes / n; Avg. = unweighted mean of the four unrounded rates. Budgets cap all branches and verification calls. Bold: best and underline: second best within each budget (ties at the shown precision share the mark; not a significance test). Shaded: the proposed method.